BİLAL CANKARDAŞ

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PROFESSIONAL SUMMARY

Senior Electronics and Communication Engineering student at Yıldız Technical University and R&D engineering intern at BİSavunma, designing a 5.8 GHz direction-finding antenna array. Experienced in antenna design and EM simulation (Ansys HFSS, CST), passive direction finding and embedded wireless systems, from simulation to field testing, focused on RF for unmanned platforms.

EDUCATION

Yıldız Technical University *October 2021 – Expected January 2027*
B.Sc. in Electronics and Communication Engineering

EXPERIENCE

R&D Engineering Intern · BİSavunma *September 2026 – Present*

- Designing a 5.8 GHz four-element circular array (UCA) for angle-of-arrival direction finding in Ansys HFSS; printed monopole element on Rogers RO4003C with S11 of −33.2 dB (VSWR 1.08) and 3.02 dBi realized gain.
- Chose 0.40–0.45 λ spacing to avoid λ/2 bearing ambiguity (45°/225°); next step is PCB layout in KiCad.

Antenna & Microwave Intern · ARASA Lab, Yıldız Technical University *July 2026 – August 2026*

- X-band horn with a WR-90 coax-to-waveguide transition (CST); a 393-run sweep kept S11 < −20 dB over 8–12 GHz.
- 2.4 GHz inset-fed patch on FR-4 (HFSS, MATLAB dimensioning): S11 ≈ −29 dB, 2.72 dBi; 75-page HFSS tutorial.

ACADEMIC & TECHNICAL PROJECTS

Kinematic-Aware, Event-Triggered Routing for Wearable LoRa Mesh *March 2026 – June 2026*
Graduation Project (team of 3), Yıldız Technical University

- Wearable fall-detection system with pre-impact confirmation for field personnel, communicating over an 868 MHz LoRa mesh network with motion-aware, event-triggered routing; the proposed protocol achieved the highest packet delivery ratio among eight compared protocols.
- Contributed to hardware integration and field testing of the multi-node testbed, and analysed its RSSI measurements to assess link quality under motion and body shadowing.

Vivaldi Antenna and SDR Receiver for Electronic Support *October 2025 – January 2026*
Design Project (team of 3), Yıldız Technical University

- Researched SDR receivers and passive direction-finding methods (RSSI, interferometry, MUSIC, TDoA/FDoA) for the team's 0.8–6 GHz Vivaldi antenna; simulated RSSI bearing search and 2-element MUSIC in MATLAB.

Radar Cross Section Reduction of a Coated Missile *December 2025 – January 2026*
Term Project (team of 3), Electronic Defense Systems, Yıldız Technical University

- Modeled a 1.9 m missile in CST with absorber coatings on the nose, body and fins for a bare-vs-coated monostatic RCS comparison; reviewed shaping, RAM and cancellation methods.

Wearable Patient Monitor over Bluetooth Low Energy *November 2025 – January 2026*
Medical Electronics and Embedded Systems Project (team of 2), Yıldız Technical University

- nRF52840 device that detects falls, immobility and distress sounds and reads heart rate and SpO2 (MAX30100, I2C), streamed over a custom BLE GATT service.

Multidisciplinary Design Project — Wearable Health Monitoring System *November 2025 – January 2026*
Term Project (team), Yıldız Technical University

- Rechargeable wearable integrating pulse, SpO2 and accelerometer sensors for continuous monitoring.

SKILLS

RF & Antenna: Ansys HFSS, CST Studio Suite, antenna and array design, impedance matching, direction finding (AoA, MUSIC), RCS

Embedded & Wireless: ESP32, nRF52840 (Arduino Nano 33 BLE), I2C, SPI, UART, BLE, LoRa

Software & Tools: MATLAB, C, LTspice, OrCAD/PSpice, Logisim

Languages: Turkish (Native), English (Advanced), Japanese (A1)